



20 mm caliber

20 mm caliber is a specific size of popular autocannon ammunition. The dividing line between smaller-caliber weapons, commonly called "guns", from larger-caliber "cannons" (e.g. machine gun vs. autocannon), is conventionally taken to be the 20 mm round, the smallest caliber of autocannon. All 20 mm cartridges have an outside projectile (bullet) diameter and barrel bore diameter of approximately 0.787 inches (20.0 mm). These projectiles are typically 75 to 127 mm (3–5 in) long, cartridge cases are typically 75 to 152 mm (3–6 in) long, and most are shells, with an explosive payload and detonating fuze.

Weapons using this caliber range from anti-materiel rifles and anti-tank rifles to aircraft autocannons and anti-aircraft guns.

Usage

Twenty millimeter caliber weapons are generally not used to target individual soldiers, but rather objects such as vehicles, buildings, or aircraft.

Types of ammunition

- High explosive (HE)
- High explosive incendiary (HEI)
- Armor-piercing (AP)
- Semi-armor-piercing high explosive incendiary (SAPHEI)
- Armour-piercing discarding sabot (APDS)
- High explosive fragmentation tracer (HEF-T)
- High explosive high capacity (HE-M)
- Penetrator with enhanced lateral effect (PELE)
- Target practice - inert projectile (i.e., PGU-27A/B)^[1]
Used for training (TP)
- Target practice tracer - inert projectile with tracer material in base for visual trajectory tracking (i.e., PGU-30A/B) (TP-T)



A 20×102 mm round (second from left) with .50 BMG rounds, golf ball, and a stick of 168-pin SDRAM computer memory

20 mm weapons

Each weapon is listed with its cartridge type appended.

Current weapons

Weapon	Country of origin	Cartridge	Notes
<u>Neopup PAW-20</u>	 <u>South Africa</u>	<u>20×42 mm</u>	
<u>Denel NTW-20</u>	 <u>South Africa</u>	<u>20×82 mm Mauser</u>	
<u>Denel Vektor GA-1</u>	 <u>South Africa</u>		
<u>Vidhwansak</u>	 <u>India</u>		
<u>Anzio 20mm rifle</u>	 <u>United States</u>	<u>20×102 mm</u>	
<u>Arash anti-materiel rifle</u>	 <u>Iran</u>		
<u>M61 Vulcan</u>	 <u>United States</u>		
<u>M197 electric cannon</u>	 <u>United States</u>		
<u>M39 cannon</u>	 <u>United States</u>		
<u>GIAT M621</u>	 <u>France</u>		
<u>ZVI PL-20 Plamen</u>	 <u>Czech Republic</u>		
<u>Nexter Narwhal 20A</u>	 <u>France</u>		Based on the <u>M621</u>
<u>Şahi 20-102</u>	 <u>Turkey</u>		[2]
<u>XM301</u>	 <u>United States</u>		
<u>Denel NTW-20</u>	 <u>South Africa</u>	<u>20×110 mm Hispano</u>	
<u>Metallic RT-20</u>	 <u>Croatia</u>		
<u>Truelo CMS 20x110mm</u>	 <u>South Africa</u>		
<u>Yugoimport-SDPR M71/08</u>	 <u>Serbia</u>		Single barrel development of <u>Zastava M55</u>
<u>Oerlikon KAE (KAA/KAB)</u>	 <u>Switzerland</u>	<u>20×128 mm</u>	Formerly known as Oerlikon 204GK/5TG
<u>Meroka CIWS</u>	 <u>Spain</u>		
<u>Oerlikon KAD</u>	 <u>Switzerland</u>	<u>20×139 mm</u>	Formerly known as Hispano-Suiza HS.820
<u>GIAT M693/20 mm modèle F2</u>	 <u>France</u>		
<u>Rheinmetall Rh 202</u>	 <u>Germany</u>		
<u>Denel Land Systems GI-2</u>	 <u>South Africa</u>		
<u>Nexter Narwhal 20B</u>	 <u>France</u>		Based on the <u>20 mm F2</u>

Historical weapons

Weapon	Country of origin	Cartridge	Notes
<u>Becker Type M2 20 mm cannon</u>	 <u>German Empire</u>	20×70 mm RB	
<u>Oerlikon FF</u>	 <u>Switzerland</u>	20×72 mm RB	
<u>Type 99 cannon, model 1</u>	 <u>Japan</u>		
<u>Ikaria-Werke Berlin MG FF/M cannon</u>	 <u>Nazi Germany</u>	20×80 mm RB	
<u>Mauser MG 151/20</u>	 <u>Nazi Germany</u>	20×82 mm	
<u>Ho-5 cannon</u>	 <u>Japanese Empire</u>	20×94 mm	
<u>Berezin B-20</u>	 <u>Soviet Union</u>	20×99 mm R ^[3]	
<u>ShVAK</u>	 <u>Soviet Union</u>		
<u>Helenius RK-20 APH</u>	 <u>Finland</u>		Derivative of the <u>Helenius RK-97 12.7 mm anti-materiel rifle</u>
<u>Ghan-Krnka fortress rifle</u>	 <u>Russian Empire</u>	20.3×95 mm R	Bullet diameter 21 mm
<u>Type 99 cannon, model 2</u>	 <u>Japanese Empire</u>	20×101 mm RB	
<u>Solothurn S-18/100</u>	 <u>Switzerland</u>	20×105 mm B	
<u>Automatkanon m/45</u>	 <u>Sweden</u>	20×110 mm	Aircraft <u>autocannon</u>
<u>Automatkanon m/49</u>	 <u>Sweden</u>		Aircraft <u>autocannon</u>
<u>Hispano-Suiza HS.404 and derivatives</u>	 <u>Switzerland</u>		
<u>Oerlikon F, FFL</u>	 <u>Switzerland</u>	20×110 mm RB	
<u>Polsten</u>	 <u>Poland</u>	20×110 mm RB	
<u>Colt Mk 12 cannon</u>	 <u>United States</u>	20×110 mm USN	Advanced derivative of the HS.404
<u>Madsen 20 mm anti-aircraft cannon</u>	 <u>Denmark</u>	20×120 mm	
<u>Type 97 automatic cannon</u>	 <u>Japanese Empire</u>	20×124 mm	
<u>Mauser MG 213</u>	 <u>Nazi Germany</u>	20×135 mm	
<u>Solothurn S-18/1000 anti-tank rifle</u>	 <u>Switzerland</u>	20×138 mm B	
<u>Solothurn S-18/1100 anti-tank rifle</u>	 <u>Switzerland</u>		
<u>ST-5 20 mm AA gun</u>	 <u>Switzerland</u>		
<u>FlaK 30 and FlaK 38 single-barrel AA</u>	 <u>Nazi Germany</u>		Forerunner of Hispano-Suiza HS.820's post-war 20×139 mm round
<u>Flakvierling quadruple-barrel AA</u>	 <u>Nazi Germany</u>		
<u>MG C/30L aircraft gun</u>	 <u>Nazi Germany</u>		
<u>Rheinmetall KwK 30 and KwK 38</u>	 <u>Nazi Germany</u>		
<u>Cannone-Mitragliera da 20/65 modello 35 (Breda)</u>	 <u>Kingdom of Italy</u>		

<u>Cannone-Mitragliera da 20/77 (Scotti)</u>	 <u>Kingdom of Italy</u>		
<u>Lahti L-39 anti-tank rifle</u>	 <u>Finland</u>		<u>Solothurn Long</u>
<u>Lahti L-40 anti-aircraft gun</u>	 <u>Finland</u>		
<u>Nkm wz.38 FK anti-tank vehicle-mounted gun</u>	 <u>Poland</u>		Designated "heaviest machine gun", or " <i>Najcięższy karabin maszynowy, Nkm</i> ", AA/AT (actually an autocannon)
<u>Type 98 20 mm AA machine cannon and Type 4 20 mm twin AA machine cannon</u>	 <u>Japan</u>	20×142 mm	
<u>Automatkanon m/40</u>	 <u>Sweden</u>	20×145 mm R	<u>Autocannon</u>
<u>Pansarvärnsgevär m/42</u>	 <u>Sweden</u>	20×180 mm R	<u>Recoilless anti-tank rifle</u>

Cartridges

(incomplete list)

Dimensions	Name	Date	Bullet diameter	Case length	Rim	Base	Shoulder	Neck	Cartridge length
<u>20.3x95R</u> <u>Ghan</u>		21 (.827)	95 (3.74)	-	-	-	-	-	
<u>20x150R</u>		~21 (~.827)	150 (5.9)	-	-	-	-	-	

Naming conventions

The usual nomenclature of ammunition indicates the diameter of projectile and the length of the cartridge that holds it; for example, 20×102 mm is a 20 mm projectile in a 102 mm long case.

Though this designation is often assumed to be unique, this is not always the case, e.g. there are three different 20×110 mm types which are not compatible. These may be distinguished in that some cartridge designations may include additional letters or names as a suffix, e.g. the various different types of 20×110 mm might be distinguished as *20×110 mm Hispano*, *20×110 mm RB* and *20×110 mm USN*.

Common suffixes

- **B** e.g. 20×138*B*: the cartridge has a belt which is used for headspacing, i.e. it helps ensure the correct positioning within the gun's chamber.
- **R** e.g. 20×145*R*: a rimmed cartridge: the diameter of the rim forming the base is larger than that of the cartridge case itself.
- **RB** e.g. Oerlikon 20×110*RB*: rebated rim, one where the rim is a smaller diameter than the case head allowing the extractor to follow it into the chamber, facilitating advanced-primer ignition, a recoil-moderating system.

References

1. PGU-27A/B TP/ PGU-28A/B SAPHEI / PGU-30A/B TP-T (<http://www.globalsecurity.org/military/systems/munitions/pgu-28.htm>)
2. "Otomatik Orta Kalibre Top" (<https://urbansavunma.com.tr/sahi-20/>). 17 October 2022. Retrieved 17 October 2022.
3. Williams, Anthony G (January 4, 2021). "Ammunition Data Tables - 20 mm Calibre Cartridges" (<https://web.archive.org/web/20220330003338/http://www.quarryhs.co.uk/ammotable2.html>). *Military Guns & Ammunition*. Solo Publications. Archived from the original (<http://www.quarryhs.co.uk/ammotable2.html>) on March 30, 2022. Retrieved April 17, 2022.

External links

- FAS: 20 mm Cannon Ammunition (<http://www.fas.org/man/dod-101/sys/land/20mm.htm>)
- ATK produced 20, 25 & 30 mm caliber ammunition (http://www.atk.com/customer_solutions_internationalproducts/cs_ip_ma_m-c.asp)
- Rapid Fire: 20 mm Cartridge Data Table (<https://web.archive.org/web/20141109220010/http://www.quarryhs.co.uk/ammotable2.htm>)
- Rapid Fire: 20 mm Antitank Rifle Cartridges image (<http://www.quarryhs.co.uk/ATR2.jpg>)
- Rapid Fire: 20 mm Autocannon Cartridges WWII image 1 (<http://www.quarryhs.co.uk/20mm1.jpg>)
- Rapid Fire: 20 mm Autocannon Cartridges WWII image 2 (<http://www.quarryhs.co.uk/20mm2.jpg>)
- Rapid Fire: 20 mm Autocannon Cartridges Post-WWII image (<http://www.quarryhs.co.uk/20mm3.jpg>)

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